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MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: B.Tech. (Aero)

Semester: V

Name of the Course: Aircraft Stability and Control

Course Code: TAS 504

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer **all the questions** by choosing **any one of the sub questions**.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO 2
(a)	Define static stability of an aircraft. With the help of neat sketches, explain the different types of static stability (stable, unstable, and neutral). Discuss the physical significance of a restoring moment in ensuring equilibrium flight.	
OR		
(b)	Differentiate between static stability and dynamic stability of an aircraft. Explain, with suitable sketches, how positive damping and negative damping affect the time response of an aircraft when disturbed from its equilibrium position. Also comment on the role of a stability augmentation system (SAS) in improving dynamic stability.	
Q2	(10 marks)	CO3
(a)	Consider two airplanes with linear pitching moment curves as shown in Figure 1. Both are trimmed at the same equilibrium point. (a) Explain why an airplane with a negative slope of the pitching moment coefficient curve with respect to angle of attack exhibits longitudinal static stability. (b) Discuss why an airplane with a positive slope of the pitching moment curve becomes statically unstable when subjected to an angle-of-attack disturbance. (c) Illustrate the importance of trimming capability at a positive angle of attack and describe the condition on the pitching moment coefficient at zero lift (C_{m_0})	

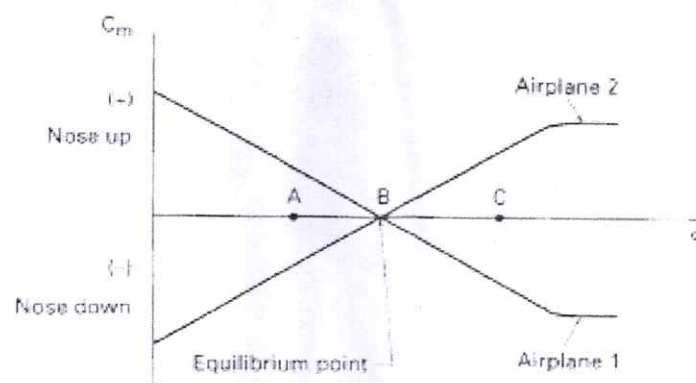


	Figure 1	
	OR	
(b)	The longitudinal static stability of an airplane can be analyzed using the pitching moment coefficient curve either as a function of angle of attack (α) or lift-coefficient(C_L). (a) Derive the relationship between $\frac{dC_m}{d\alpha}$ and $\frac{dC_m}{dC_L}$ and explain its physical significance. (b) Why is it important for aircraft designers to evaluate the individual contributions of the wing, fuselage, tail, and propulsion system to overall static stability? (c) Discuss how systematic methods such as the USAF Stability and Control Dotcom assist in preliminary aircraft design for ensuring stability across different flight regimes.	
Q3	(10 marks)	
(a)	Derive the simplified expression for the pitching moment coefficient of the wing about the center of gravity, starting from the general moment equation and stating the assumptions for small angles of attack.	CO3
	OR	
(b)	Derive the governing equation for the contribution of the horizontal tail to the longitudinal static stability of an aircraft. In your derivation, clearly indicate the assumptions made and express the resulting relation in terms of stability derivatives. Support your answer with a neat and properly labeled sketch illustrating the geometry of the horizontal tail with respect to the aircraft reference line and center of gravity.	
Q4	(10 marks)	
(a)	The wing–fuselage pitching-moment characteristics of a high-wing, single-engine, general-aviation airplane is given about the center of gravity by $C_{m_{cg_{wf}}} = -0.05 - 0.0035\alpha$ where α is the fuselage-reference-line angle of attack (in degrees) and the subscript wf denotes the combined wing–fuselage contribution. The pertinent geometric and aerodynamic data are: $S_w = 178 \text{ ft}^2, b_w = 35.9 \text{ ft}, AR_w = 7.3, \frac{c_w}{c} = 5.0 \text{ ft}, \frac{x_{cg}}{c} = 0.10, i_w = 2.0^\circ, C_{L_{0_{wf}}} = 0.26, C_{L_{\alpha_{wf}}} = \frac{0.07}{\text{deg}}$	CO4

	Design the horizontal tail (planform area and incidence) so that the complete airplane (wing–fuselage–tail) exhibits the following pitching-moment characteristics about the center of gravity: $C_{m_{cg_{wft}}} = 0.15 - 0.025\alpha$ where the subscript wft denotes the combined wing–fuselage–horizontal-tail contribution and α is in degrees. Assume the following data for the horizontal tail: $l_t = 15 \text{ ft}, \eta = 1, AR_t = 5, C_{L_{\alpha_t}} = 0.073/\text{deg}$ Determine: - the required horizontal tail area S_t, and - the required tail incidence angles i_t.	
	OR	
(b)	Derive the expressions for the change in lift and pitching moment due to elevator deflection. Define elevator effectiveness and elevator control power, and explain how design parameters such as tail volume ratio and flap size affect elevator control authority.	
Q5	(10 marks)	
(a)	With the help of a neat sketch, explain the primary aerodynamic control surfaces of an aircraft (elevator, rudder, aileron, and spoiler). Discuss briefly how each surface contributes to pitch, yaw, and roll control of the airplane.	CO3
	OR	
(b)	(a) Derive the relationship between stick force and the hinge moment of the elevator. (b) With the aid of a neat sketch, explain the function of a trim tab and discuss how it helps in eliminating stick forces at trimmed flight conditions.	

